

Research papers

LSTM time series NDVI prediction method incorporating climate elements: A case study of Yellow River Basin, China

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ABSTRACT

Accurate prediction of the trend of Normalized Difference Vegetation Index (NDVI) time series in the Yellow River Basin (YRB) is crucial for the assessment of the hydrological and ecological environment in this region. Currently, the NDVI time series prediction model is primarily based on traditional models and single-variable neural network models. Nevertheless, these models present challenges in considering the limitations of multiple factors, causing the NDVI time series prediction results to lack reliability. To predict NDVI time-series in the YRB of China, this study constructed a multilayer multivariate Long-Short Term Memory (LSTM) neural network model including climatic components. The initial important climatic elements in this region were identified using GeoDetector. Then, the relationship between NDVI and climatic factors in the YRB of China is established. Finally, numerical scale data are used to train and predict a multilayer multivariate LSTM model with climatic components. According to the results, the three-layer multivariate LSTM neural network NDVI time series prediction model developed in this study has the best performance among the evaluated indices. When compared to existing time series prediction models, the proposed model in this study takes into account the common constraint effect of various climate factors on NDVI. This leads to a significantly improved prediction accuracy, presenting new opportunities for enhancing the prediction model. By analyzing the NDVI time series prediction outcomes for the YRB, it has been determined that the ecological environment of the area will continuously improve in the future. This study offers significant technological and theoretical backing for assessing the hydrological and ecological environment of the YRB and comparable ecologically vulnerable regions in China.

1. Introduction

There exists a complex and interrelated link between hydrological processes and ecosystems (Gou et al., 2023). A comprehensive comprehension of this interaction is vital for the sustainable management of local water resources, the preservation of the health of ecological environments, and the adaptation to climate change (Pradhananga and Pomeroy, 2022; Chen et al., 2023a). The hydrological process entails numerous elements, including rainfall, evaporation, transpiration, and runoff. Ecosystems influence hydrological processes through plant transpiration, soil water retention, and water cycling (Xie et al., 2023; Pesántez et al., 2023). The most significant interdependence between the two is the effect on water resources (Pradhananga and Pomeroy, 2022; Pesántez et al., 2023). Consequently, the influence of hydrological

processes on ecosystems is often portrayed through their effect on vegetation cover (Fang and Pomeroy, 2023). On this basis, the NDVI is considered the most useful indicator for measuring the coverage and growth status of surface vegetation in a given region (Xu et al., 2022). This contributes to a greater understanding of the link between hydrological processes and ecosystems. The monitoring and prediction of long-term NDVI changes can provide insights into the processes and future trends of vegetation change (Zhang et al., 2023a; Fu et al., 2022). This information facilitates a timely understanding of the vegetation cover of the regional hydrological and ecological environment, which in turn can help to develop appropriate measures to protect the environment and manage water resources (Rahman et al., 2022).

Located at the crossroads of arid, semi-arid and semi-humid regions in China, the YRB serves as an important ecological barrier and corridor

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(Tian et al., 2021). The region's water resources exhibit an imbalanced supply and demand and low self-regulation capacity. The hydrological and ecological environment is extremely fragile with a slow and challenging recovery process (Du et al., 2023). Research indicates considerable greening and browning of vegetation in the YRB during recent years (Tian et al., 2021). Different driving factors contribute to and create spatial heterogeneity in vegetation growth (Ren et al., 2022; Zuo et al., 2023). However, there is minimal research on NDVI temporal prediction, especially for particular regions and numerical scales (Zhang et al., 2021b). Therefore, it is imperative to thoroughly examine the spatiotemporal correlation of NDVI in the YRB to achieve more precise predictions for the future trends of vegetation change (Gao et al., 2023). This is fundamental for enhancing the comprehension of the interplay between hydrological processes and ecosystems (Chen et al., 2023b). Not only does it provide a scientific basis for protecting the hydrological and ecological environment of the YRB, but it also helps to promote regional ecological balance and environmental sustainability more effectively.

The current research on time series prediction predominantly utilizes conventional statistical techniques, including AutoRegressive Integrated Moving Average (ARIMA), Holt-Winters, Markov chain analysis, and so on (Wang et al., 2023; Shin et al., 2023). However, the usage of these methods has limitations in terms of nonlinear fitting ability and model structure, which results in inadequate accuracy in time series prediction (Kim et al., 2023; Cao et al., 2023b). Machine learning has been extensively utilized in time series prediction research (Zhang et al., 2023b; Zhou et al., 2023) as it can extract possible non-linear and complex information from time series data. The same holds for the spatiotemporal prediction of NDVI (Khoshkalam et al., 2023; Fei et al., 2023; Ahmad et al., 2023). As an instance, Stepchenko and Chizhov (2015) employed Elman Recurrent Neural Networks (ERNN) in NDVI time series prediction studies and displayed that more precise univariate prediction models can be developed using neural networks. Wang et al. (2022a) utilized LSTM models, which were driven by water levels in different sub-regions, to evaluate the effect of diverse ecological water recharge scenarios on both spatial and temporal changes of NDVI in wetlands. Cui et al. (2020) suggested Scale-Free Convolutional Neural Network (SF-CNN) models to enhance the precision of NDVI prediction, displaying robust generalization capability, and not necessitating other data co-modelling. Nevertheless, NDVI is limited by various climatic factors (Lehner et al., 2023). Current deep learning temporal prediction models only take into account the time-related characteristics of NDVI, and hardly ever consider the constraining implications of climate factors (Gong et al., 2021; Reddy and Prasad, 2018). Furthermore, establishing the internal connection between the past and future NDVI (Gao et al., 2023) remains a challenging task that has resulted in poor NDVI time series prediction outcomes. It is paramount to predict plant growth trends accurately in China's YRB in the future by incorporating climatic factors.

The research area for this study is the YRB of China. The purpose of the study was to develop an LSTM neural network NDVI time series prediction model that integrates climatic parameters to forecast vegetation patterns in the region in the future. This study aims to (1) screen the primary climatic influences on vegetation NDVI in the YRB of China using GeoDetector based on the time-series process of vegetation NDVI changes and climatic influences, and (2) develop a multi-layer multi-variate LSTM time-series NDVI prediction model based on the primary climatic influences to achieve time-series prediction of vegetation NDVI in the YRB of China. This study provides theoretical and data support for ecological conservation and improved quality in the YRB of China.

2. Materials

2.1. Study area

The Yellow River of China rises in the Bayankara Mountains and

flows through nine provinces. Bordered by the Bohai Sea in the east, the Qinling Mountains in the south and the Yin Mountains in the north, the YRB (106°-112°E, 34°-40°N) covers a total area of 794,500 km², accounting for about 8.3 % of China's land area (Fig. 1). The YRB is an important security barrier and an ecological and economic corridor in China. The terrain is high in the west and low in the east, spanning China's three necessary terrain levels with diverse topography and landforms. The YRB is located in the mid-latitude zone with a continental climate and marked climatic differences. There are wide temperature variations between seasons and between years, with cold, dry weather in winter and humid, rainy summers with concentrated precipitation and high evapotranspiration (Chao et al., 2023). The natural vegetation of the YRB is influenced by the marine monsoon, with forest-steppe, steppe and desert-steppe zones appearing from south-east to north-west, forming a variety of ecosystem types, with steppe, desert and irrigated agricultural regions dominating in the upper reaches, steppe, forest and dry farming regions in the middle reaches, and dry farming and wet areas in the lower reaches (Zheng et al., 2023). The average NDVI values in the YRB from 2000 to 2021 are mainly in the range of 0.21–0.52, with an insignificant increasing tendency.

2.2. Data sources and preprocessing

Data from the MOD13A2 product is derived from the gridded secondary consequence of the MODIS Terra satellite, which provides NDVI data at 1 km spatial resolution every 16 days and provides quality assurance (QA) to ensure data quality (Cao et al., 2021). In this study, we used the MOD13A2 product from LAADS DAAC (<https://ladsweb.modaps.eosdis.nasa.gov/>) to acquire NDVI data from 2000 to 2021. We used MODIS Reprojection Tool (MRT) software for batch-implement processing such as geometric correction, projection stitching and resampling. Subsequently, to reduce the influence of clouds on the data, we used the Maximum Value Composite (MVC) method to synthesize the data after the above processing into monthly NDVI data, which were cropped (Holben, 1986). Also, to reduce the influence of soil background and to enhance the information of NDVI data, we removed areas with NDVI values less than 0 throughout the study period and extracted the characteristic attribute values.

The characteristics of the data used in this study are shown in Table 1. Meteorological data was obtained from the National Science and Technology Infrastructure Platform-National Earth System Science Data Sharing Service Platform-Land Plateau Science Data Centre (<https://loess.geodata.cn>). Land Surface Temperature (LST) data was obtained from NASA's (<https://earthdata.nasa.gov/>) MODIS surface temperature product MOD11C3, which provides monthly 0.05-degree latitude/longitude Climate Modelling Grid (CMG) surface temperature and emissivity (LST&E) values. Potential Evapotranspiration (ET₀) data was obtained from the National Tibetan Plateau Science Data Centre (<https://data.tpdc.ac.cn/home>) at a spatial resolution of approximately 1 km and were calculated using the Hargreaves potential evapotranspiration formula. To meet the needs of the study, the above data were formatted, converted to coordinates and cropped to obtain monthly value raster data from 2000 to 2021, and we used feature extraction techniques to extract the feature values.

As the dynamic vegetation cover is relatively stable in the short term, to deal with the MODIS NDVI time series data gaps. In this study, we used linear interpolation to fill the data and the Savitzky-Golay filter to eliminate the filtering disturbances (Wang et al., 2022b). At the same time, the presence of extreme values in the data, such as temperature and precipitation, makes the data take on an extensive and unevenly distributed range of values, increasing the complexity of the neural network and affecting the prediction outcomes. Normalizing the data and compressing data attribute values between [0,1] is necessary. Preserving necessary features of the data, reducing the importance of parameters, and eliminating the impact of dimensionality and order of magnitude can be useful tactics. This approach can also speed up the

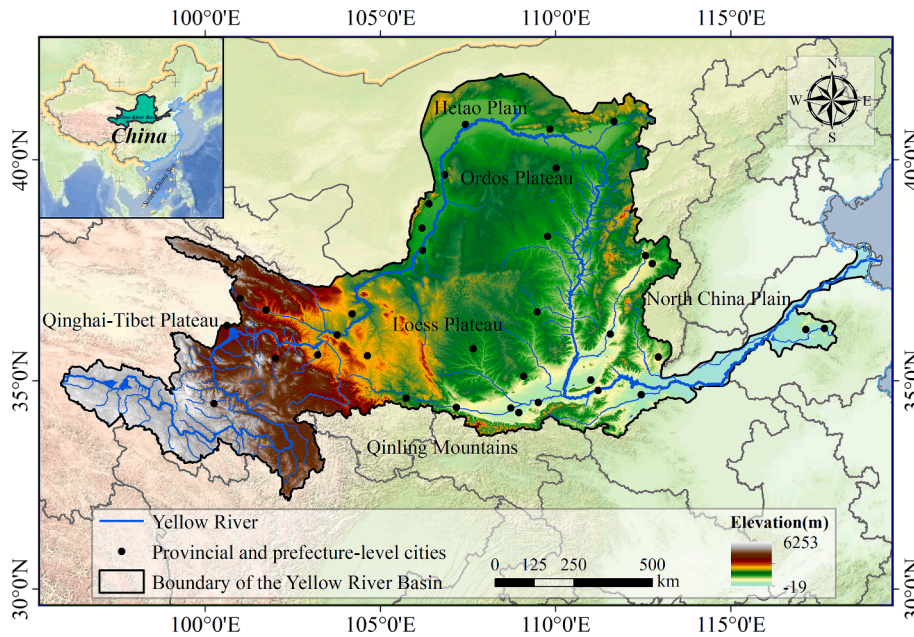


Fig. 1. Geographical location map of the YRB, China.

Table 1

Characteristics of the data used in this study.

Product	Spatial resolution	Temporal resolution	Timespan used in model development	Data usage
MODIS NDVI (MOD13A2)	1 km	16 days	2000–2021	Forecast
Meteorology data	0.05°	monthly	2000–2021	Ancillary
MODIS LST (MOD11C3)	0.05°	monthly	2000–2021	Ancillary
ET ₀	1 km	yearly	2000–2021	Ancillary

convergence of neural networks and enhance prediction efficiency.

3. Methods

In this study, we first used GeoDetector to screen climate elements with high NDVI importance and establish links between them and NDVI to capture spatiotemporal correlations. We then constructed a multi-layer multivariate LSTM time-series prediction model incorporating climate elements for predicting future NDVI trends and compared it with a univariate neural network model. The technical flowchart is shown in Fig. 2.

3.1. Statistical methods

3.1.1. GeoDetector

The study employed GeoDetector to screen for climatic factors that are of high importance in relation to NDVI. GeoDetector is a statistical technique used to detect spatial variability and identify the key drivers affecting the spatial heterogeneity of a dependent variable. The fundamental premise is that if an independent variable substantially affects a dependent variable, then the spatial distribution trends of both variables should be similar (Wang and Xu, 2017; Zhang et al., 2021a). In multi-factor correlation analysis, GeoDetector displays greater explanatory power and analytical precision compared to the traditional correlation coefficient method (Gong et al., 2022).

3.1.2. Pearson correlation coefficient

The Pearson correlation coefficient measures the closeness between two or more correlated variables. It is usually expressed in terms of PCCs, which range from [-1,1], with absolute values closer to 1 indicating a stronger correlation between the variables (Qiu et al., 2020; Guo et al., 2021). Thus, we quantified the degree of correlation between NDVI and selected climate elements by PCCs to better understand the spatial and temporal patterns between them.

3.1.3. Trend analysis

In this study, we used the integrated linear trend method and the Mann-Kendall (MK) test to analyze the trends in NDVI at different stages. The linear trend method is based on the continuity of time development principle, comparing the same indicators or ratios in different data periods and directly observing the increase or decrease in changes and the magnitude of changes to examine the trend (Qiu et al., 2019; Li et al., 2023; Cao et al., 2023a). It is suitable for any sample distribution and can effectively avoid the influence of a few outliers on the test results (De Jong et al., 2011).

3.1.4. Residual analysis

To assess the accuracy of the NDVI predictions, we used residual analysis to check the differences between the predicted and actual results to evaluate the quality of the model fit. Residual analysis is a statistical method that assesses the fit of a statistical model by comparing the difference between the actual and predicted values (Cao et al., 2022).

3.2. NDVI time series prediction model

3.2.1. ARIMA model

The ARIMA model is a commonly used tool for analysing and forecasting time-series data (Yan et al., 2022). ARIMA is a particularly useful approach for handling time-series data that displays trends, seasonality, or randomness (Zhang et al., 2023c). The primary idea behind ARIMA is to transform time-series data into a stationary process by using the differencing technique, and then applying autoregressive (AR) and moving average (MA) models to explain its characteristics (Wang et al., 2023; Shin et al., 2023; Yan et al., 2022).

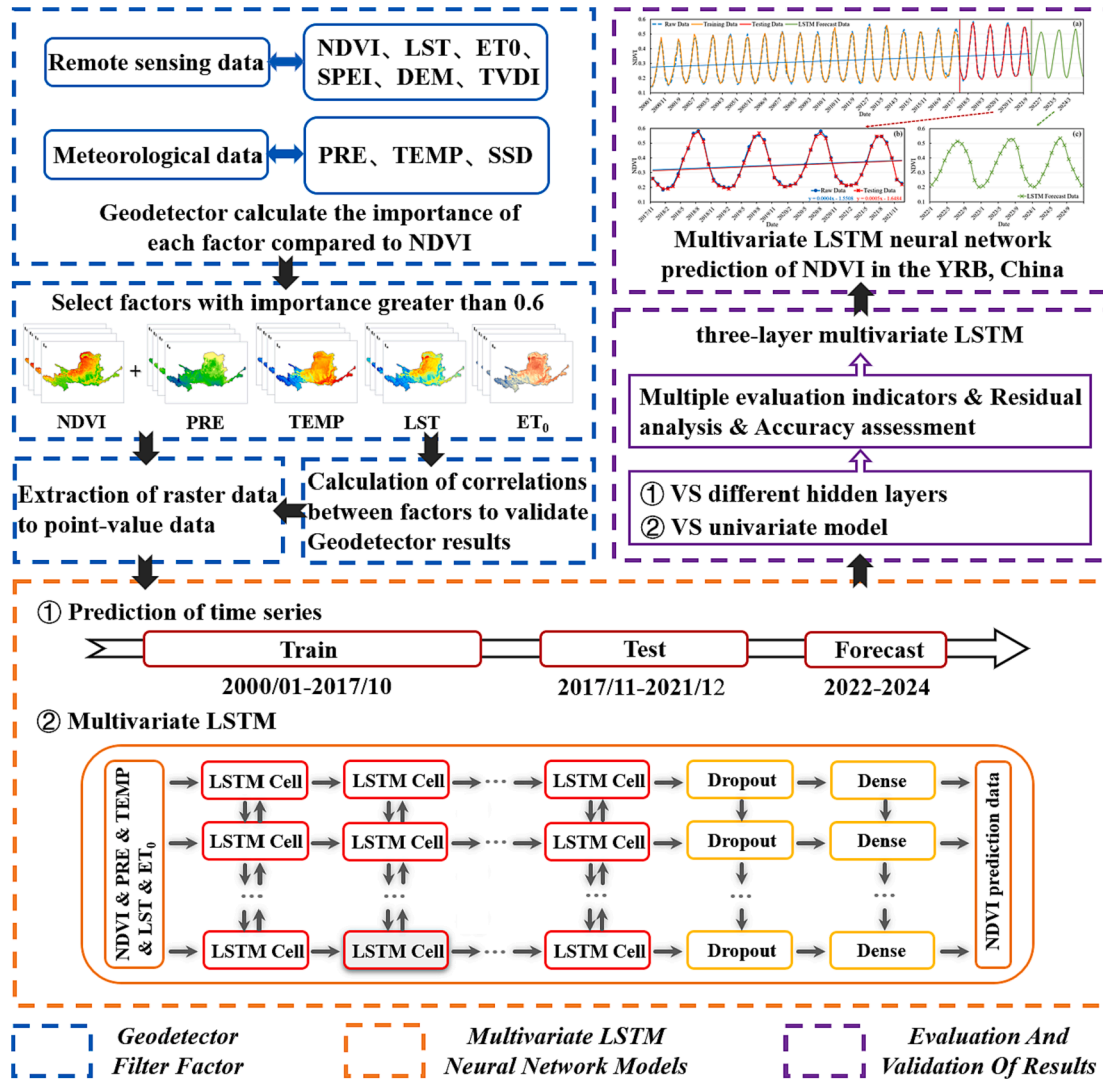


Fig. 2. The technical flowchart of this study.

3.2.2. Holt-Winters model

The Holt-Winters model is a widely accepted technique for forecasting time series data with trend and seasonal characteristics (Shin et al., 2023). It consists of three main components: Namely trend, seasonality, and smoothing. The central concept behind the technique is to forecast future data points based on historical observations of trends, seasonality, and smoothing information (Kim et al., 2023).

3.2.3. RNN model

The Recurrent Neural Network (RNN) is a type of neural network that can iteratively process sequential information (Yang et al., 2022). It has a cyclic structure that permits the neurons to be connected in a loop, thereby preserving past temporal information for the present and influencing the output. This structure facilitates neurons to establish interconnections and maintain information about the previous time sequence while considering the current input for impacting the output (Shen et al., 2020). Nevertheless, when processing extensive time series, the RNN encounters difficulties in maintaining the critical information from the past, affecting its effectiveness (He et al., 2021). To address this issue, adaptive memory or deliberately forgetting of the information can be achieved by incorporating gating mechanisms (Wang et al., 2022a; Chen et al., 2023a).

3.2.4. LSTM neural network model

LSTM is an RNN option initially proposed in 1979 (Hochreiter and Schmidhuber, 1979). LSTM inherits most features of RNN models, making it well-suited to handle problems closely related to time series. LSTM can better solve the issues of gradient explosion and gradient dispersion as compared to traditional RNN models (He et al., 2022; Zhao et al., 2022b). Multiple layers of the network gradually forget the preceding information over time. Thus, LSTM can describe or simulate human behavior, logical evolution, and the cognitive processes of neurons more accurately (He et al., 2021; Courtial et al., 2022). LSTM can store long-term sequence information and is suitable for simulating problems with multiple input variables (Chen et al., 2021).

The core idea of the LSTM is to use gating mechanisms to determine whether and how the LSTM layers are updated using gates to update cell states (He et al., 2023; Chen et al., 2021). Each LSTM layer (Fig. 3(c)) contains one cell state and three neural network layers (forget, input, and output gate layer). Updating the information at the current time t determines how much information is retained from time $t-1$, whether the input information at time t is updated, and controls how the output is produced (Zheng et al., 2019; Zhang et al., 2021a). The LSTM at time t is calculated as follows:

$$\text{Forgetgate layer} : f_t = \sigma(W_f^T[h_{t-1}, x_t] + b_f) \quad (1)$$

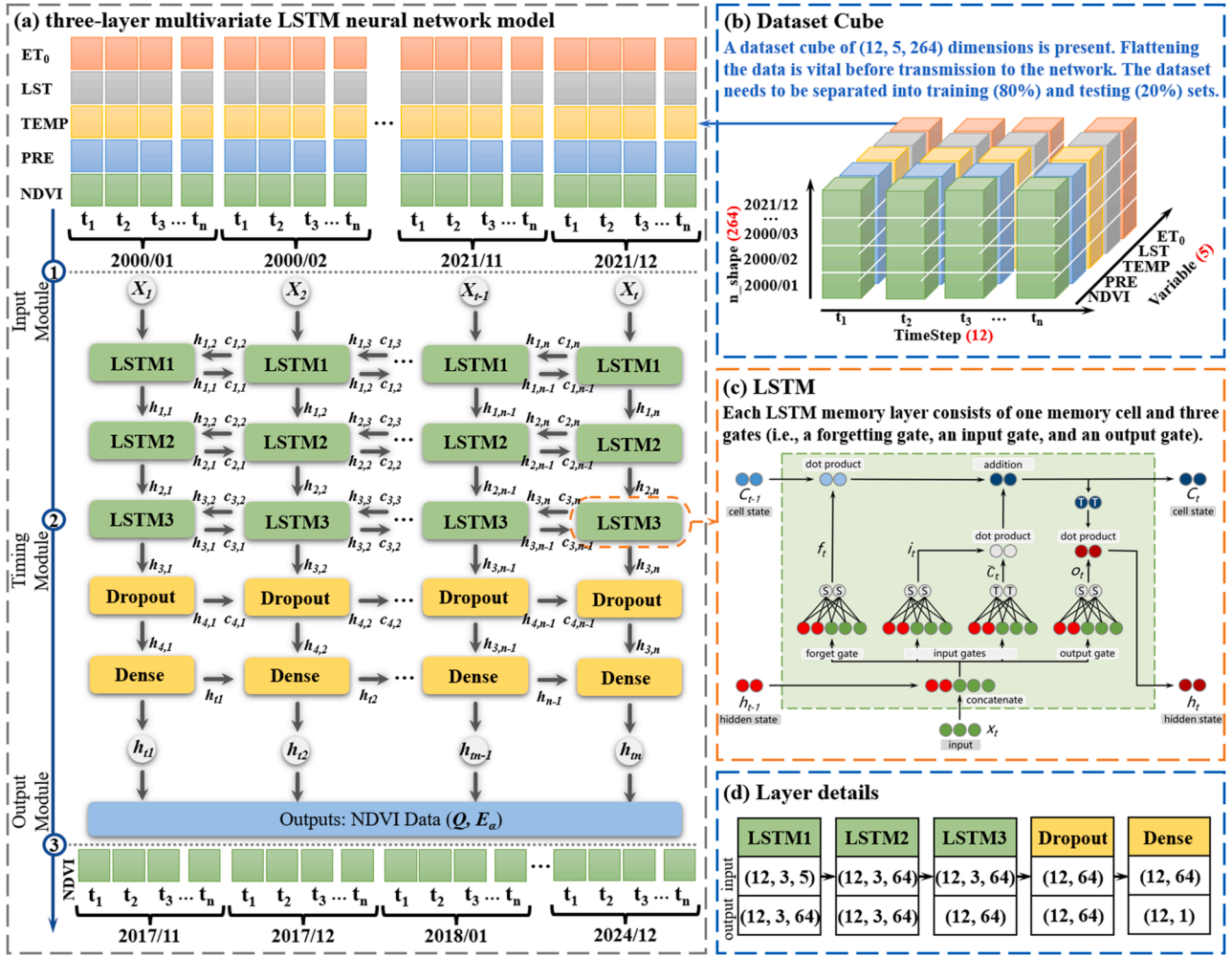


Fig. 3. (a) Overall architecture of the three-layer multivariate LSTM model for NDVI prediction in the YRB, China. (b) The number of cube dimensions in the original dataset. (c) Within LSTM cells. The meaning of each variable is explained in the text. (d) The number of data dimensions before and after each layer in this research model.

$$\text{Inputgate layer} : i_t = \sigma(W_f^\circ [h_{t-1}, x_t] + b_i) \quad (2)$$

$$\text{Outputgate layer} : o_t = \sigma(W_o^\circ [h_{t-1}, x_t] + b_o) \quad (3)$$

$$\text{Cellstate} : C_t = f_t^\circ C_{t-1} + i_t^\circ \tilde{C}_t \quad (4)$$

$$\text{Memorybody} : h_t = o_t^\circ \tanh(C_t) \quad (5)$$

$$\text{Candidatestate} : \tilde{C}_t = \tanh(W_c^\circ [h_{t-1}, x_t] + b_c) \quad (6)$$

Where, the symbol ' $\circ$ ' denote the dot product operation of 2 vectors; W_f, W_i, W_o, W_c denote the weight vectors of the forget gate layer, input gate layer, output gate layer and cell state respectively; b_f, b_i, b_o, b_c denote the bias vectors of the forget gate layer, input gate layer, output gate layer and cell state respectively; σ and $\tanh$ are the activation functions (Greff et al., 2017; Zhao et al., 2022a).

3.3. Method of constructing an NDVI time series prediction model incorporating climate elements

3.3.1. Model construction

LSTM is able to recognise features in input data (He et al., 2023; Yao et al., 2023) through training for classification, prediction or decision making. LSTM performs well in temporal prediction. Nonetheless, LSTM may face situations where error accumulation leads to inaccurate

predictions due to its univariate limitations (Jiang and Liu, 2023). To explain the complex correlations of NDVI over a long-term period, we added climate factors as constraints to our model's input section. In conclusion, we developed a multilayered and multivariate LSTM model capable of forecasting NDVI in the YRB. We consider the three-layer multivariate LSTM model architecture an example (Fig. 3(a)). The model architecture comprises an input module, a timing module, and an output module.

Notably, we construct a bidirectional interworking stacked LSTM layer in the timing module to capture complex temporal features in the input dataset (Wang et al., 2021; Wang et al., 2022a). This method can simultaneously process the input of the time series in both forward and backward directions, and participate in computing the output of the current time t with the cell state vectors generated at each time t . It then connects the forward and backward outputs and passes them to the next layer. This method uses the same LSTM layer instead of two independent LSTM layers, unlike conventional Bidirectional Long Short-Term Memory (Bi-LSTM).

The Keras neural network API in Python was used in this study to create a model for predicting NDVI in the YRB. Before feeding the data into the model, the mean and standard deviation of the dataset are calculated to scale all the data (Wang et al., 2022b). Our model is designed with multiple layers (Fig. 3(d)), and GPU is utilised to swiftly train the model, owing to the volume of input data and the complexity of the matrix (He et al., 2021). A Dropout layer was added to prevent

overfitting. The Dense layer was implemented to achieve complex non-linear mapping and feature extraction (Chen et al., 2021; Yao et al., 2023). To perform a grid traversal search, the GridSearchCV class in scikit-learn was employed since distinct parameters combinations have a substantial effect on prediction accuracy.

The dataset cube is flattened into a matrix (Fig. 3(b)), with 80 % as the training set and the remaining 20 % as the testing set. TimeStep is set to 3. All hyperparameter combinations were thoroughly explored, and the optimal parameters were determined, achieving an accuracy of about 77.62 %. The study concluded by adopting the Adam optimisation algorithm, with a learning rate of 0.001, momentum of 0.09, a batch_size of 16, and 100 epochs. Simultaneously, we use triple cross-validation (CV) to confirm the dependability of the best parameters (Chen et al., 2023a; Zhao et al., 2022b).

3.3.2. Model evaluation

To assess the efficacy of the model set out in this article, we have developed multiple evaluation approaches. First, we compared the efficiency of multivariate LSTM models with distinct hidden layers for predicting the NDVI time series. Secondly, we employed different traditional and advanced learning models like ARIMA, Holt-Winters, and RNN models to predict the NDVI time series. In addition, we calculate some standard metrics to evaluate the performance of different models, including Coefficient of determination (R^2), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), Mean Decrease Accuracy (MDA) and Explained Variance Score (EVS), which help to judge the accuracy of the prediction results (Zhao et al., 2021; Zhao et al., 2022a; Feng et al., 2022; Blaskowitz and Herwartz, 2011; Sinclair et al., 2010; Ramraj et al., 2018). They can be obtained by the following methods:

$$R^2 = \frac{\sum_{t=1}^n (\hat{y}_t - \bar{y})^2}{\sum_{t=1}^n (y_t - \bar{y})^2} \quad (7)$$

$$MAE = \frac{1}{n} \sum_{t=1}^n |\hat{y}_t - y_t| \quad (8)$$

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{\hat{y}_t - y_t}{y_t} \right| \times 100\% \quad (9)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (10)$$

$$MDA = \frac{1}{n} \sum_{t=1}^n 1_{\text{sign}(y_t - y_{t-1}) = -\text{sign}(\hat{y}_t - \hat{y}_{t-1})} \quad (11)$$

$$EVS = 1 - \frac{\text{Var}\{y - \hat{y}\}}{\text{Var}\{y\}} \quad (12)$$

Where y_t is the mean NDVI value at time t ; 1 is the indicator function in equation (11); The Var is defined as the second-order moments of the probability density function in equation (12).

4. Results and discussion

4.1. Screening of factors influencing NDVI in the YRB

Multi-factor interactions can improve the interpretation of the temporal distribution and variability of NDVI (Wang et al., 2019). Due to the complexity of the NDVI prediction problem, it involves multiple influencing factors (Zhao et al., 2021). Therefore, we need to consider multiple factors when building a multivariate LSTM prediction model. The importance here refers to the interactive synergistic relationship between factor-based variables. Still, it can only represent, to a certain extent, the degree of correlation between the dependent and

independent variables. The results of GeoDetector are presented in Fig. 4. The importance of all factors in this study is higher than 20 %. Still, to make the cumulative loss error in the prediction process of the model smaller, in this research, we only select factor-type variables with importance higher than 60 %, namely PRE, TEMP, LST and ET₀. This shows that precipitation and air temperature are the main factors affecting NDVI in the YRB, and the interaction between precipitation and NDVI is more significant (Zhang et al., 2021b). In addition, LST is significantly negatively correlated with NDVI changes (Yan et al., 2010), while ET₀ enhances the interpretation of NDVI spatial distribution in a non-linear manner (Jiang et al., 2022). In turn, it can be shown that the climatic factors screened by GeoDetector are all physically well-interpreted and establish a robust interactive correlation in the YRB.

To better assess the response relationship between each factor and NDVI, we calculated the PCCs between them (Table 2) and plotted a data distribution plot (Fig. 5). The data show a good linear fit and the direction of slope change is obvious. The data is evenly distributed on both sides of the trend line and shows a positive influence. NDVI varied with the other factors, especially the best correlation between precipitation and NDVI, with PCCs reaching 0.9155 and the flattest confidence ellipse. This indicates that climatic factors can easily affect the NDVI of the YRB. This phenomenon may be due to the fact that the YRB belongs to arid and semi-arid regions and is sensitive to changes in precipitation and temperature, making it more responsive to the effects of climate change. The correlations between the other factors also showed promising results, with the corresponding PCCs above 0.76 and in general agreement with the results calculated by the GeoDetector, proving their results are reliable.

4.2. Time series prediction of NDVI in the YRB incorporating meteorological factors

Univariate prediction methods use a single factor as input data to make a prediction. Yet, the use of univariate forecasting methods fails to offer an all-inclusive depiction of problem characteristics (Kiani Shahvandi and Soja, 2022). To elevate the accuracy of predictions, this study presents multiple periodic factors to establish multivariate forecasting models. Crucial to time series predictions is the decomposition and reconstruction of datasets. A data set for a supervised model can be built using a sliding window, enabling the multivariate prediction model to assimilate and grasp the attributes of the primary time series data, including the relative position and length of values in the cycle (Bhoj and Singh Bhadoria, 2022; Malakouti, 2023; Song et al., 2023). Iterative computation is carried out to validate and adjust predictions and make way for the estimation of future values.

In this study, NDVI data from the YRB were used as the basis. The

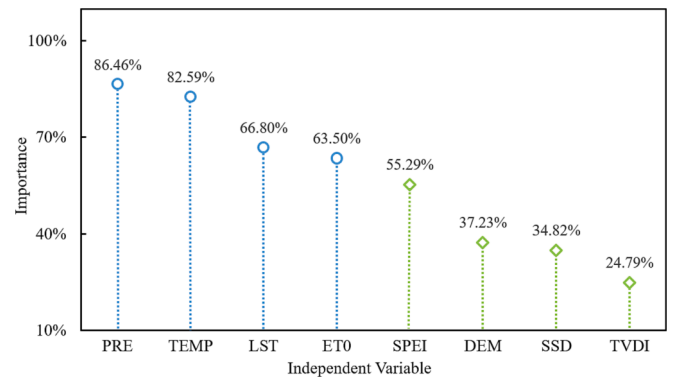


Fig. 4. Vertical plot of the importance of climate factors screened by the GeoDetector. The blue indicates the factor-based variables selected for this study, while the green colour is unselected. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
PCCs between NDVI and each factor.

Variable	NDVI	TEMP	PRE	LST	ET ₀
NDVI	1.0000	—	—	—	—
TEMP	0.8822	1.0000	—	—	—
PRE	0.9155	0.8479	1.0000	—	—
LST	0.7895	0.9804	0.7650	1.0000	—
ET ₀	0.7973	0.9653	0.7805	0.9701	1.0000

relevant climatic factors were introduced on the basis of their factors, selected solely on the basis of their relevant mean values. LSTM neural network models were utilized to establish time series prediction models, incorporating various hidden layers of data. To train the models, the initial 80 % of the NDVI monthly mean data set, covering 264 months from 2000 to 2021, was utilized, with the latter 20 % being designated for data validation by predicting information post-November 2017. The comparison of NDVI time series prediction outcomes from various

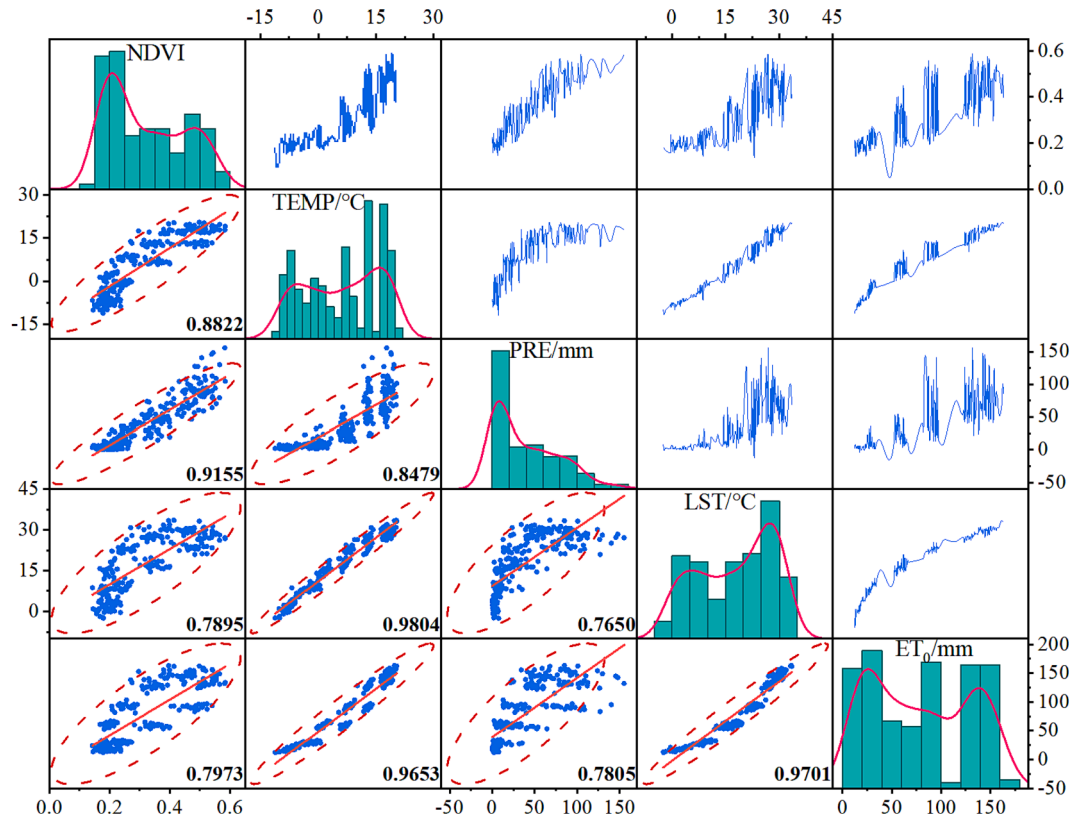


Fig. 5. Scatter matrix of the distribution of NDVI concerning each factor. The lower left corner is a scatter density plot, which determines the degree of data concentration based on the flatness of the confidence ellipse and the slope of the trend line, and displays the PCC values; The diagonal represents the statistical histogram of the data distribution; The upper right corner is the data trend distribution line plot.

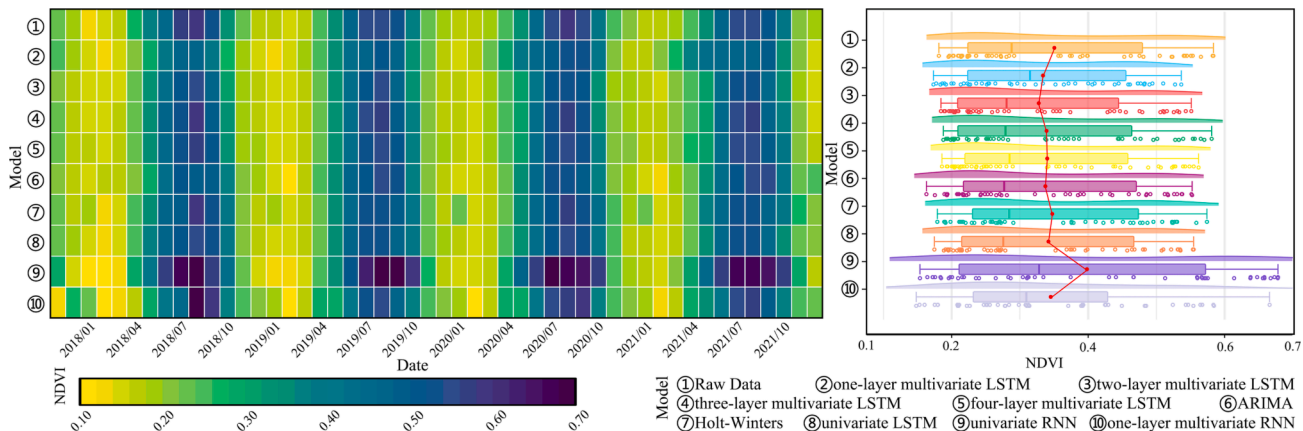


Fig. 6. Overall distribution of predicted and actual data for each model in the prediction validation phase. The heat map on the left side shows the specific numerical magnitude of the forecast results arranged in a matrix of 50 10 and mapped to a green and blue gradient given a pattern, showing the monthly value forecast results for each model on the time series on the rows and a comparison of the forecast results for different models for a given month on the columns. The statistical distribution of each model's data prediction outcomes between 0.1 and 0.7 is shown on the right-hand side as a peak range plot, with the height and size of the wave showing the degree of concentration of the data and the red dash indicating the change in the overall median of the data. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

neural network models is presented. Moreover, a comparison is made with results from traditional time series prediction models and RNN models.

4.2.1. Timing prediction results

This study compares the variability between the models' NDVI predictions and the original data by analyzing the colour changes during the prediction validation phase. This is shown in Fig. 6. The original data exhibited wave distribution mostly within 0.1 and 0.7. However, the univariate RNN model had the least well-fitted prediction results, being almost uniformly distributed between 0.1 and 0.7. The wave distribution of the remaining models was largely similar, with little variation in the concentration of the density distribution. Comparing the models, the three-layer multivariate LSTM model was found to be the most similar to the original data in terms of peaks and data distribution, suggesting that it was the better model for predicting NDVI in the YRB. The data indicate a strong interannual variability, where the mean NDVI in YRB remains at about 0.5 in summer and approximately 0.25 in other seasons. The univariate RNN model presents the most significant difference in data distribution from the original dataset. In contrast, the distinct models fail to determine the precise distribution in the heat map, attributable to similar colour shades. On the other hand, the multilayer multivariate LSTM model demonstrated considerably higher prediction stability.

4.2.2. Comparison of prediction results for different implied layer models

We plotted comparative line plots and scatter density plots to further analyze multivariate LSTM models' prediction accuracy with different numbers of implied layers (Fig. 7 and Fig. 8). By comparing the fluctuations in the prediction results of the models, we found that the prediction results of the models were similar. The multivariate LSTM models with different numbers of implied layers could accurately predict the cyclical variation of NDVI. They matched the actuality without large displacement deviations, while the prediction curves were smooth and flat. The predictions of the peaks and troughs were entirely accurate, with the predicted values ranging from 0.1 to 0.6. Moreover, the scattered distribution of the actual and expected values of the four models is very concentrated. The trend line equation coefficients are all higher than 0.8, indicating that the multivariate LSTM prediction models are feasible and can predict the overall trend of NDVI with near

accuracy, showing the advantages of multivariate LSTM models in terms of long-term memory capability. In addition, the general NDVI predictions of the two-, three- and four-layer multivariate LSTM models are mostly slightly lower than the measured NDVI values, with RMSE values of 1.6971, 1.2469 and 1.3352, respectively. In contrast, the one-layer multivariate LSTM model has approximately the same number of predictions below the valid values as above the actual values but with an RMSE of. Among them, the predicted values of the three-layer multivariate LSTM model are closer to the fundamental values than the other prediction models, with a trend line equation coefficient of 0.9863. All the model evaluation indicators are the best in their respective evaluation criteria, with R^2 as high as 0.8967, MDA and EVS of 0.9387 and 0.8985, respectively, indicating that the prediction accuracy of the three-layer multivariate LSTM model is the network structure is complete and stable and reliable.

4.2.3. Comparison of different model prediction results

The prediction performance of different time series prediction models is shown in Fig. 9. It can be observed that traditional prediction models can precisely predict NDVI values, with noteworthy R^2 and RMSE values. Nevertheless, notable oscillations may arise in the trough (Fig. 9(a)(b)). The reason for this is that the NDVI dataset comprises of intricate nonlinear connections, which these models cannot adequately account for, resulting in unstable predictions (Shin et al., 2023). Simultaneously, traditional prediction models lack the capacity to handle noise and outliers.

The univariate LSTM model can predict NDVI values accurately, with an excellent fit for peaks and troughs. The fluctuation curve between the true and predicted values of the univariate RNN model differs significantly. The predicted values have exceptionally high peaks for several months, with results near 0.7. The R^2 is just 0.4247, and the RMSE is 9.2249. Furthermore, when increasing the number of hidden layers, the RNN model forecast a significant fluctuation in the curve, with decreased data flatness and a smaller deviation in monthly displacement. The R^2 is 0.6782 and RMSE is 18.8531. The predicted values are predominantly higher than the actual values. The three-layer multivariate LSTM prediction model remains the most effective, exhibiting a smooth prediction curve, high evenness, and robust peak and valley fitting, as well as superior evaluation indicators. These findings suggest

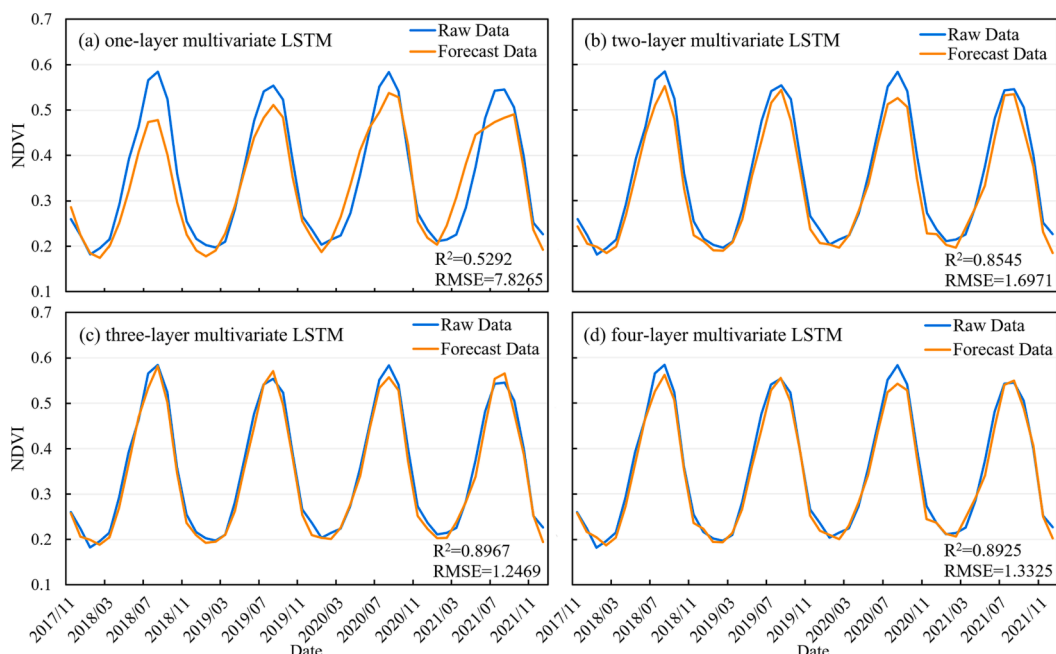


Fig. 7. Comparison line graph of prediction results of multivariate LSTM models with different numbers of implied layers.

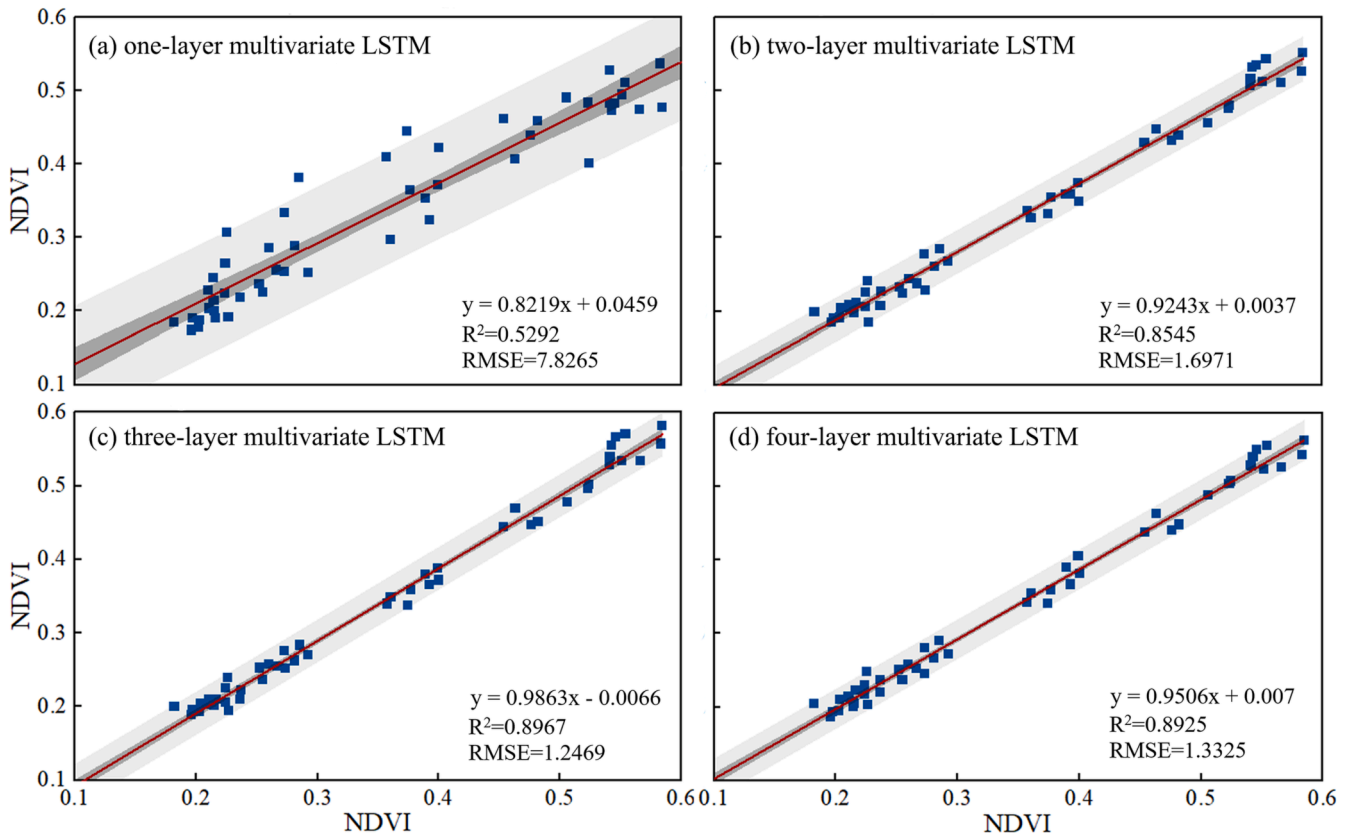


Fig. 8. Scatterplot of the predicted density of the multivariate LSTM model with different numbers of implied layers. We fitted numerous scatter points using linear trend analysis, and the trend lines all show a significant increase.

that the incorporation of periodic climate factors can enhance the predictive capabilities of multivariate models (Lehner et al., 2023). Furthermore, in comparison to other neural network models, the multivariate LSTM model with three layers possesses multiple memory units at each layer, enabling it to comprehend and learn intricate effects and relationships among multiple variables. Simultaneously, it demonstrates superior expressive power and automatically manages the respective significance of variables by learning dynamic weight allocation (He et al., 2022).

4.3. Time series prediction results in validation

According to the statistical results in Table 3, the evaluation indicators of the seven prediction models performed differently. The indicators of the three-layer multivariate LSTM model are all optimal. The RMSE is 4.1472 lower than the average of other models and the R^2 is 0.2324 higher than the average of other models. The other indicators are also superior to the average of other models to varying degrees. Because of this, it is feasible to use the three-layer multivariate LSTM model for NDVI time-series prediction in the YRB. Overall, the multivariate LSTM model with three layers yields superior prediction performance. This could be attributed to the depth of the model, which allows it to comprehensively and seamlessly capture the features of the NDVI time series prediction data cube, resulting in a highly intricate, abstract, and layered nonlinear matrix representation (He et al., 2021). Each memory unit captures feature patterns at various time scales, layer by layer. This accurately fits changes in the data and enables better learning of interaction patterns and relationships within the data (Khoshkalam et al., 2023). In addition, the multi-layer LSTM model resolves the issues of gradient vanishing and exploding through the gating mechanisms and memory units, enabling stable gradients propagation over longer sequences (Yao et al., 2023; Chen et al., 2023a). At the same time, effective

management of gradient flow assists in training more stable models.

Furthermore, we noted that the multivariate LSTM model's prediction ability progressively improved with a growing number of implicit layers, yet it lessened after including four layers. This effect can arise for the following reasons: Primarily, the structural intricacy of the multivariate LSTM model may surpass the limit that NDVI prediction can handle, which in turn can reduce the model's prediction capability. Secondly, the unstable climate in the YRB and its substantial randomness and uncertainty prevail, which, during multiple iterations, may intensify errors and accordingly lower the model's prediction results.

In this study, Fig. 10 depicts the difference in residual distribution between the actual and predicted values for each model during the prediction validation stage. Line charts in various colours represent the distribution of each difference point. For plot selection, we used models featuring a difference of ± 0.6 between each month in 2020 to gain a better understanding of performance. The results demonstrate that solely the univariate RNN model displays notable differences. The trend line exhibits some periodic correlation, with the maximum difference reaching an absolute value of 0.2. In contrast, the other models show differences with absolute values remaining under 0.13, and their distribution is more concentrated. Notably, the three-layer multivariate LSTM model demonstrates superior performance. Its line fluctuation is the most minor (indicated by the red solid line in Fig. 10), and the maximum difference is a mere 0.036. The models' prediction results exhibit strong randomness and unpredictability, and their differences lack periodicity. This highlights the characteristics of the LSTM model gating mechanism, which enables the capturing of long-term dependencies in time series and thereby enhances its prediction accuracy (Fei et al., 2023; Fang and Pomeroy, 2023). Furthermore, the models' performance in predicting NDVI time series reinforces the benefits of multivariate models over univariate counterparts.

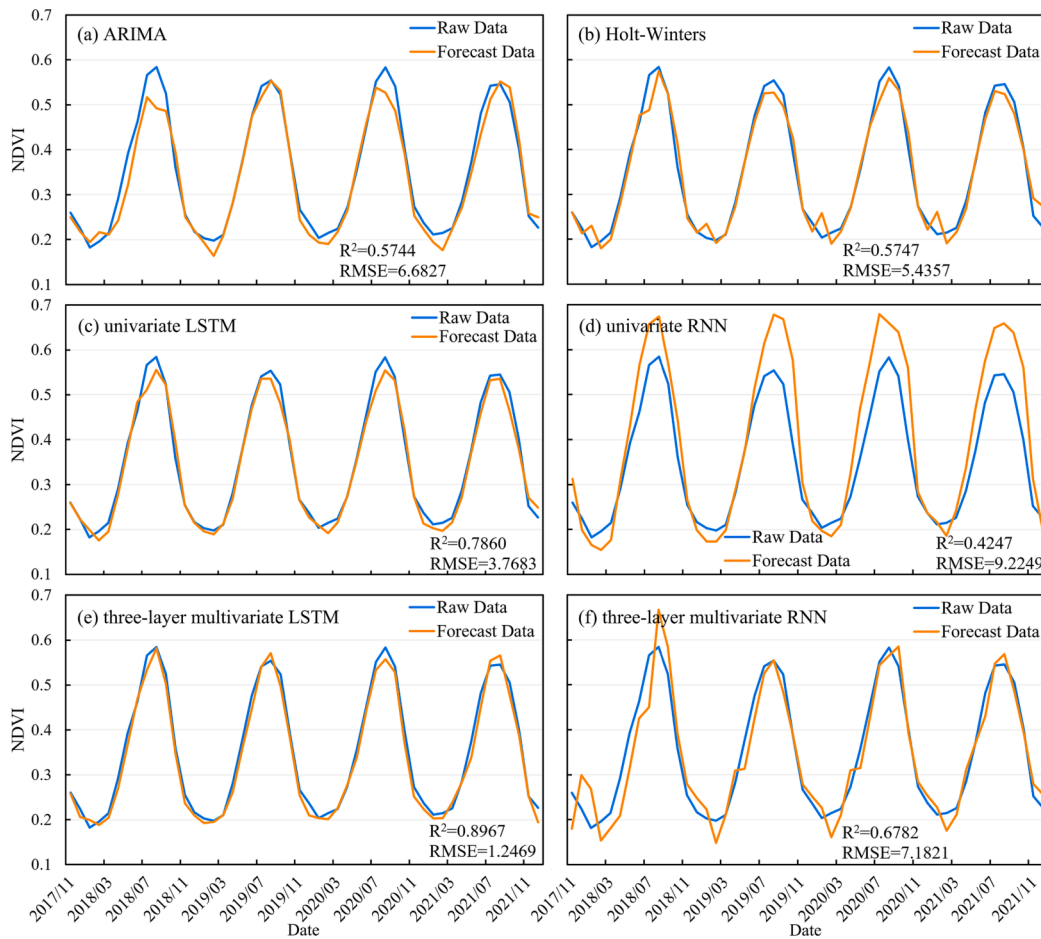


Fig. 9. Comparison of prediction effects of different time series prediction models using line charts.

Table 3
Statistical Table of Impact Index Results for Each Model Testing Stage.

Model	Evaluation R ²	MAE	MAPE	RMSE	MDA	EVS
one-layer multivariate LSTM	0.5292	8.2628	8.2361	7.8265	0.6372	0.5936
two-layer multivariate LSTM	0.8545	2.4395	6.9477	1.6971	0.9323	0.8931
three-layer multivariate LSTM	0.8967	1.2568	3.8352	1.2469	0.9387	0.8985
four-layer multivariate LSTM	0.8925	1.4241	4.2471	1.3352	0.9183	0.8979
ARIMA	0.5744	10.2733	18.9311	6.6827	0.6527	0.6225
Holt-Winters	0.5747	9.3561	15.3634	5.4357	0.6947	0.6183
univariate LSTM	0.7860	3.0226	8.6735	3.7683	0.9183	0.7861
univariate RNN	0.4247	14.9225	27.3482	9.2249	0.4856	0.4747
three-layer multivariate RNN	0.6782	5.8897	18.8531	7.1821	0.7984	0.6784

4.4. Future trends in NDVI in the YRB

According to the study results, the best-performing model for NDVI time series prediction in the YRB was the three-layer multivariate LSTM model. To ensure that the model is extensive and stable, we ran the model several times and averaged the predicted values, then

extrapolated backwards 36 months on the time series (i.e. 2022–2024). Fig. 11 shows the effect of training, testing, and forecasting the data over the full-time series. We can see that the model accurately predicts the cyclical variability of NDVI in the YRB, which is very similar to the actual data and fits well. The difference between the predicted and actual values is slight, with the maximum value not exceeding 0.02 (Table 4). The overall trend of the data in the test phase shows a significant increase, and the coefficients of the linear fit equations are nearly identical, with the two trend lines being parallel. It has been discovered that NDVI in the YRB might continue to increase in the future. The coverage rate of surface vegetation is progressively increasing, which may lead to a gradual improvement in the ecological environment of the YRB. It should be noted that this is only a predicted result and the actual situation may be influenced by many factors. Climate change, land use change and human activities can all have an impact on vegetation growth (Ahmad et al., 2023). Therefore, although the projections show positive trends, we must continue to pay attention to the development of the ecological environment. And take appropriate measures to protect and improve the ecological environment of the YRB (Li et al., 2023).

5. Conclusion

This research constructed a multi-layer, multivariate LSTM neural network to predict future trends in the surface vegetation cover of the YRB. In addition, a univariate RNN model was constructed and the test set assessment indicators and prediction results for each model were compared. The results show a strong relationship between the NDVI of the YRB and the climatic variables studied. The geographical and

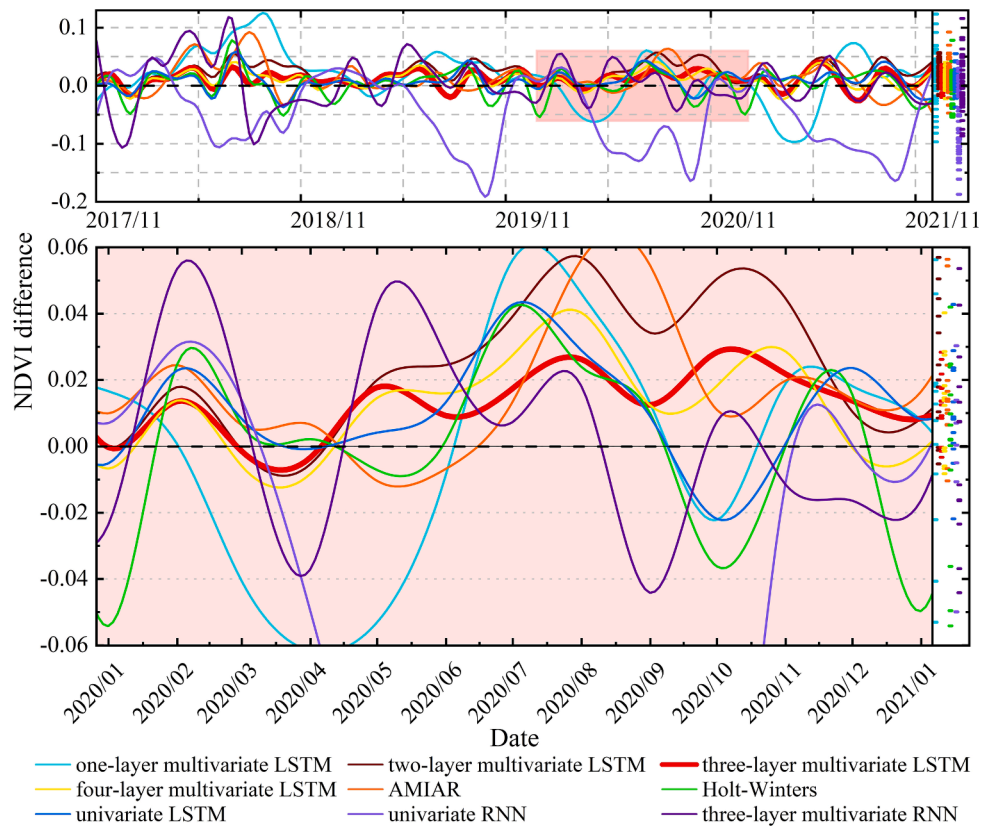


Fig. 10. Residual line chart for each stage of model prediction validation. We calculated the difference between the actual and predicted values for each model and visualized it using a line chart. At the same time, we selected each model with a difference of ± 0.6 from each month in 2020 to draw a zoom chart.

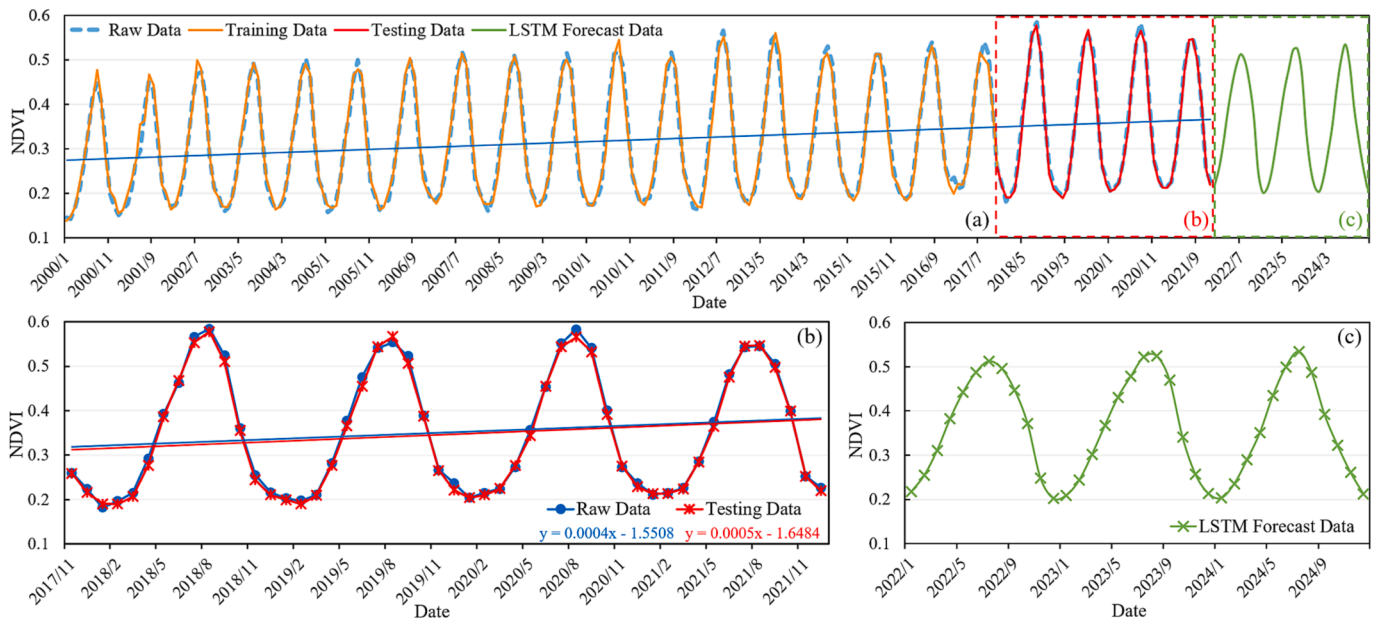


Fig. 11. Average line graphs of training, testing and prediction of NDVI time series data in the YRB. (a) is a line graph of NDVI data for the YRB from 2000 to 2024, divided into three stages: training, testing and prediction; (b) is a comparison graph of data from the raw data waiting for the 20% fitting stage; (c) is a distribution graph of data from the prediction stage.

temporal correlations of NDVI can be efficiently captured by the multi-layer multivariate LSTM-based NDVI time series prediction model, leading to more accurate forecasts. At the same time, the inclusion of periodic climatic elements improves the model's ability to predict the future. According to the best prediction model results, the biological

environment of the YRB will continue to improve.

This work validates the viability of the multi-layer multivariate LSTM model for NDVI time series prediction under climatic constraints. The NDVI time-series prediction model in this study can more accurately predict how the surface vegetation in the study region will change in the

Table 4

Statistical Table of Average Results of Impact Indicators in the Testing Stage after Multiple Runs of the Three Layer Multivariable LSTM Model.

Model	Evaluation R ²	MAE	MAPE	RMSE	MDA	EVS
NDVI Prediction Model	0.8964	1.4238	4.2729	1.2591	0.9226	0.8958

future, which is essential for determining and attributing ecological security. According to this study, the model can be extended or modified accordingly when large-scale time-series prediction of surface vegetation trend changes is required. However, the input of climatic elements needs to be more thorough, as the model only considers time series predictions at the numerical scale of monthly NDVI values. Future research will consider other regional climate variables together with spatiotemporal grid data to provide a typical NDVI prediction study in the YRB sub-basin. To achieve more accurate and comprehensive prediction results, further advances in model construction will be undertaken, including temporary module design and parameter tuning.

CRediT authorship contribution statement

Yan Guo: Conceptualization, Software, Validation, Visualization, Writing – original draft. **Lifeng Zhang:** Conceptualization, Formal analysis, Funding acquisition. **Yi He:** Methodology, Validation, Writing – review & editing. **Shengpeng Cao:** Validation. **Hongzhe Li:** Visualization. **Ling Ran:** Visualization. **Yujie Ding:** Data curation. **Mikalai Filonchik:** Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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